Tree

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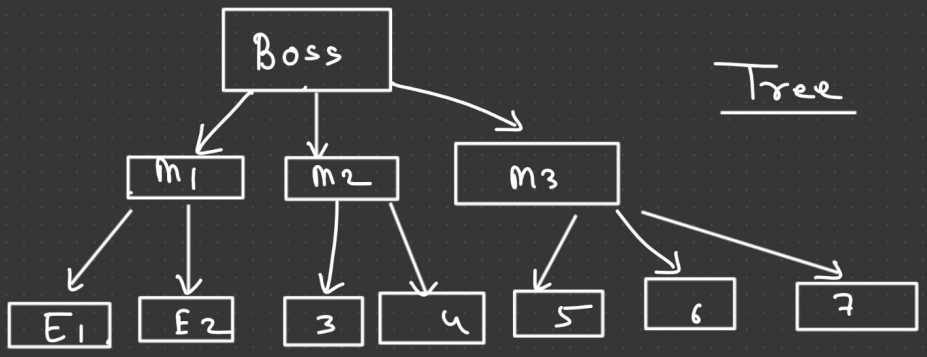
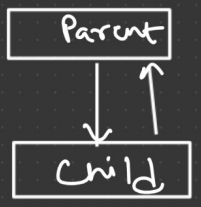
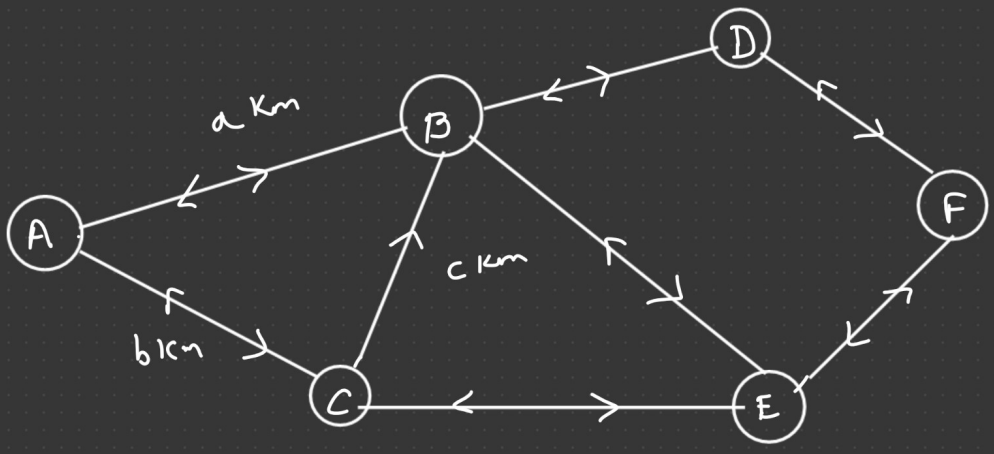
DS → 1) Linear → (Array , DA, Stack, Queue)
DS → 2) Non-Linear → (Tree , Graph)

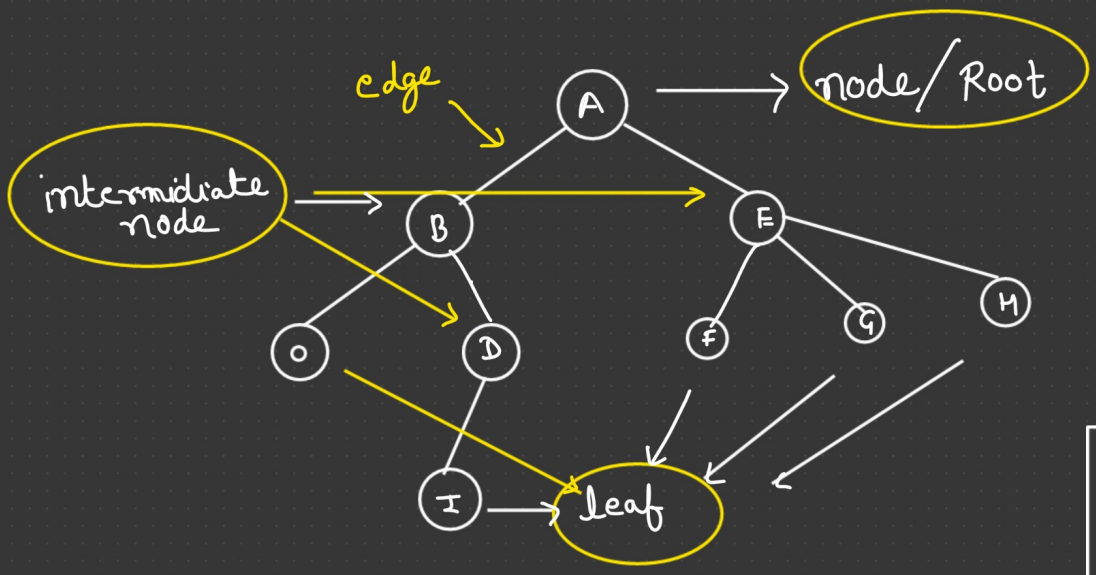


Tree





→ Graph



Root

No. of Nodes = 9
No. of Edges = 8

Leaf Node ⇒ which do not have any child node.

1) Root → first node.

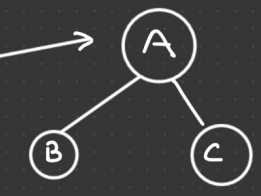
2) Leaf

3) Intermidiate Note

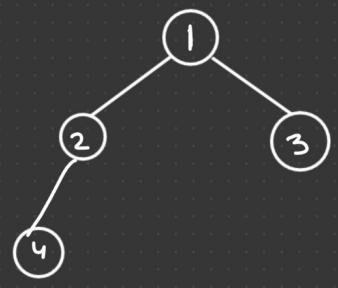
4) Edges 

5) Parent Node →

6) Child →



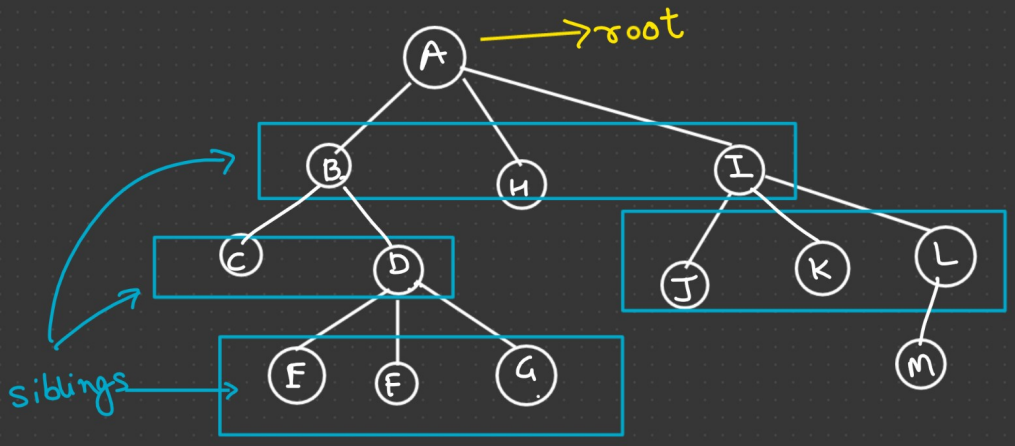
# of Nodes = 3
# of Edges = 2



★ If we have "n" Nodes in our tree then it must have "n-1" edges.

no. of edges + 1 = No. of Nodes

7) Siblings

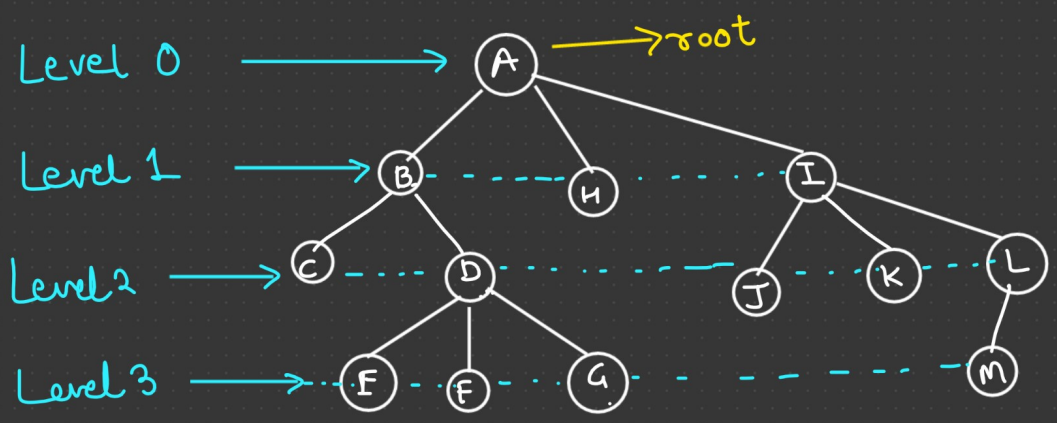


Root = A

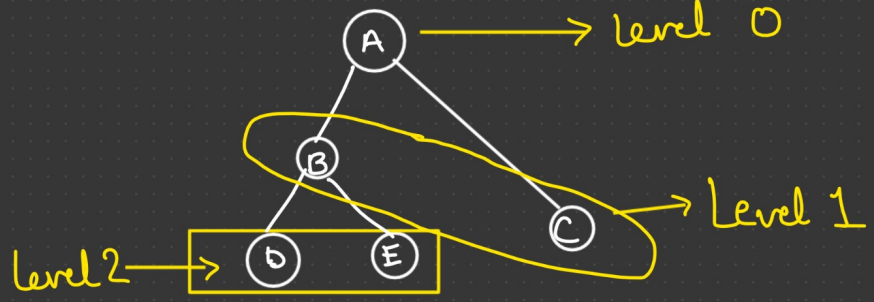
Leaf = C, E, F, G, H, J, K, M

Intermidiate = B, D, I, L

Level :-



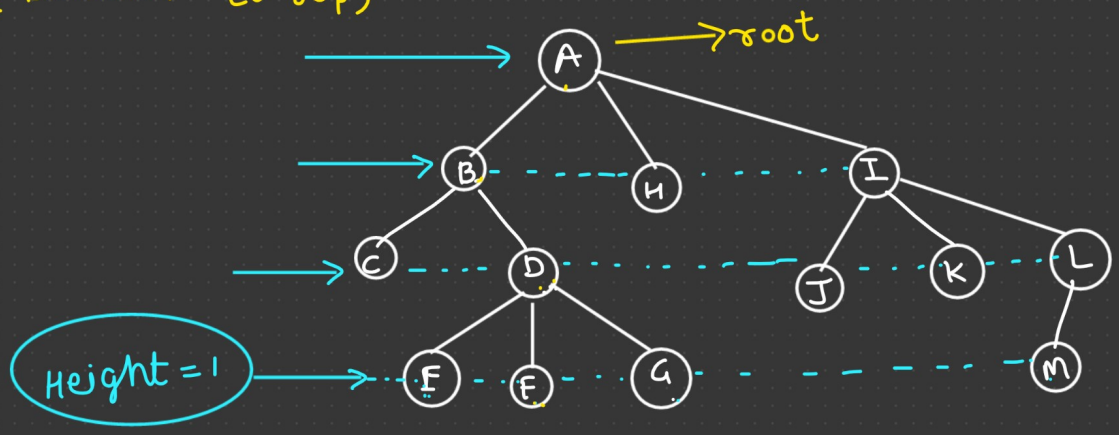
Total 4 levels → [0 - 3]



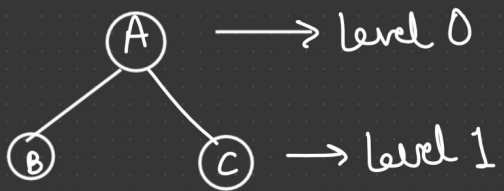
Height :- (Bottom to top)

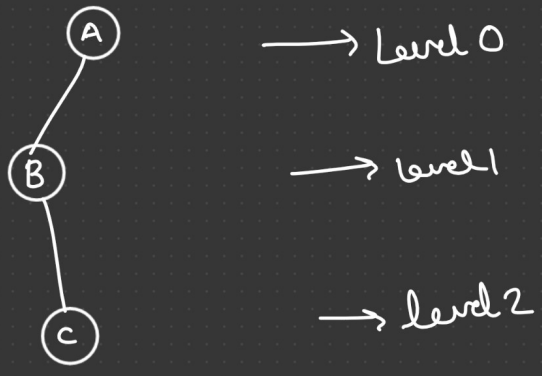
edge / node

Height = 3 / 4



Depth :- (Top to Bottom)





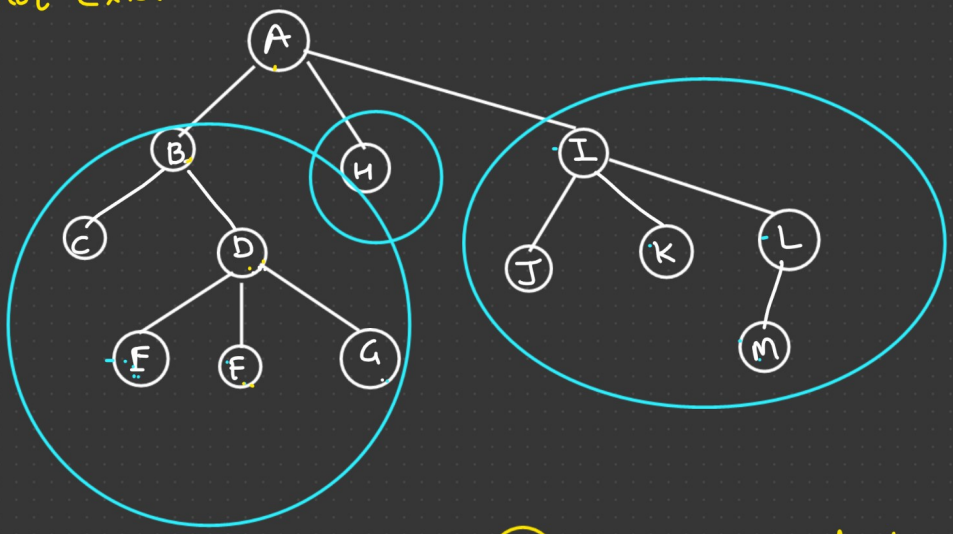
Path :-

B - K → Does not exist

A - G → A-B-D-G

A - M → A-I-L-M

B - E → B-D-E



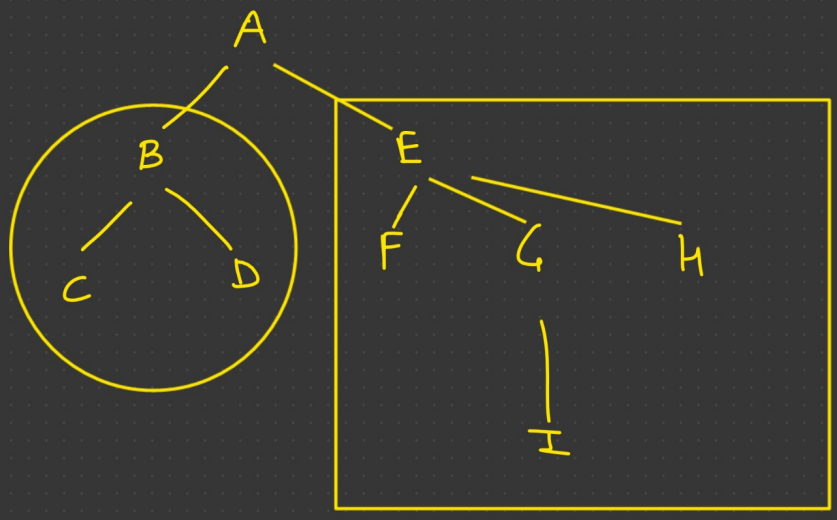
Subtree :-

$A = \{1, 2, 3, 4, 5, 6\}$

$B = \{2, 3, 6\}$

$B \subseteq A$

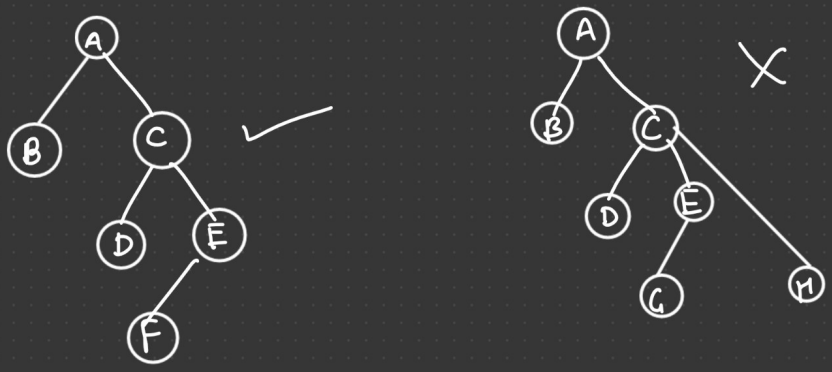
○ → single node is also a tree.



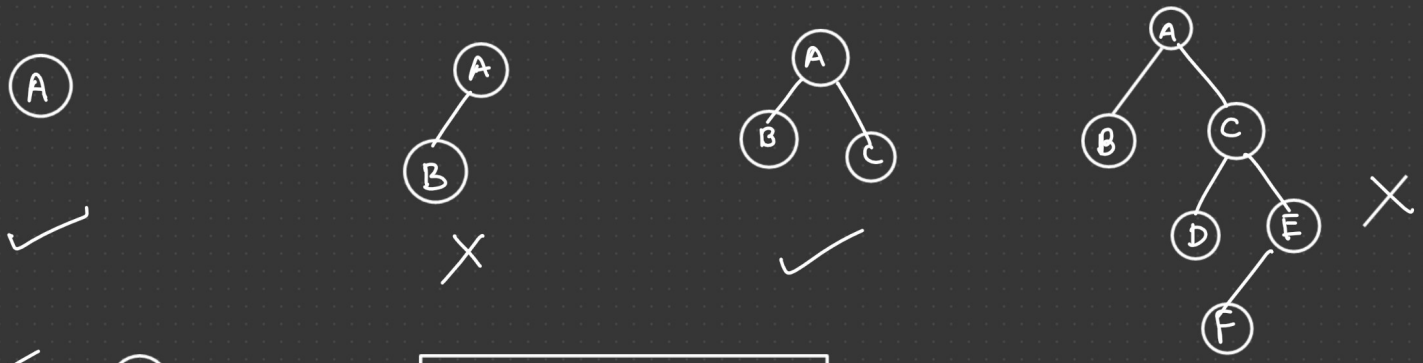
Types of Trees :-

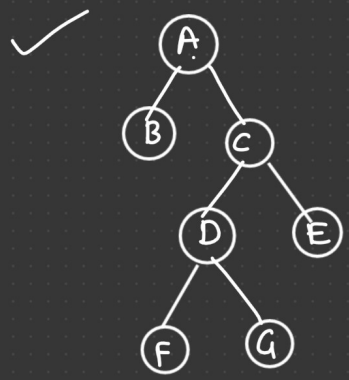
1) **Binary Tree** :- A tree is called BT if every node in it has maximum of two child nodes.

(Because it 0 child)



Strict Binary Tree :- Every node has exactly 0 or 2 child.





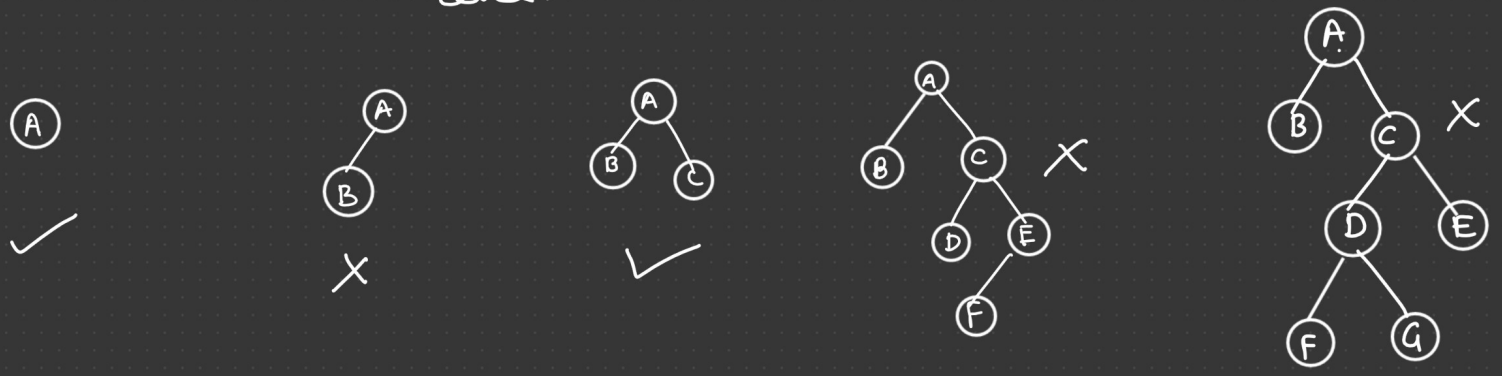
Leaf = 4

non leaf = 3

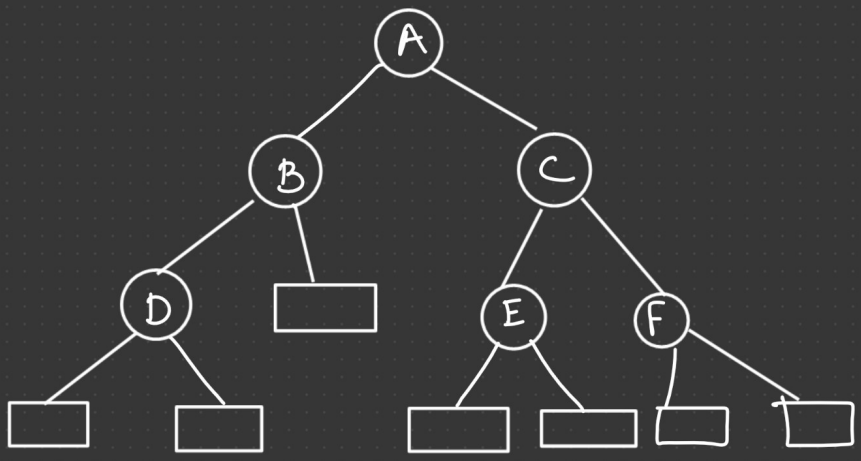
* if it has "n" non leaf node then, it must have "n+1" leaf nodes.

Complete BT :-

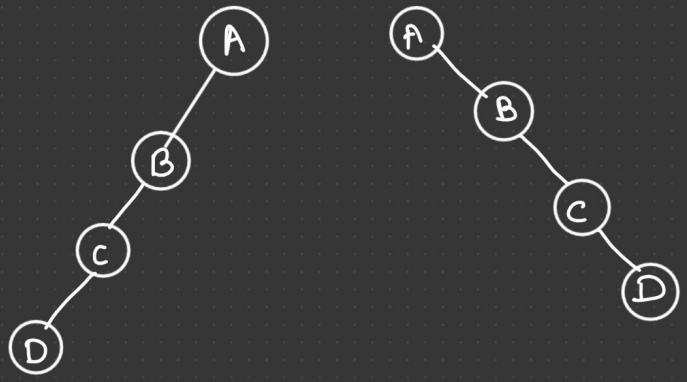
Every internal node must has exactly 2 child node. And all the leaf nodes are at the same level.



Extended BT:-



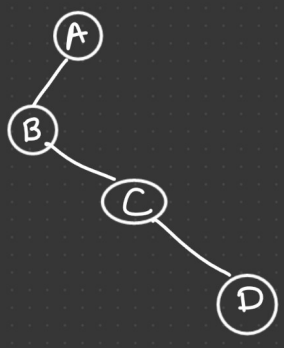
Skewed BT :-



Left - skewed

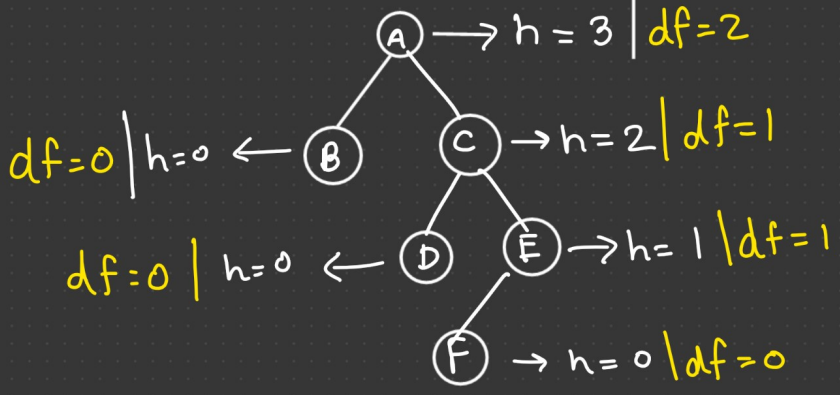
Right - skewed

Degenerate Tree :-

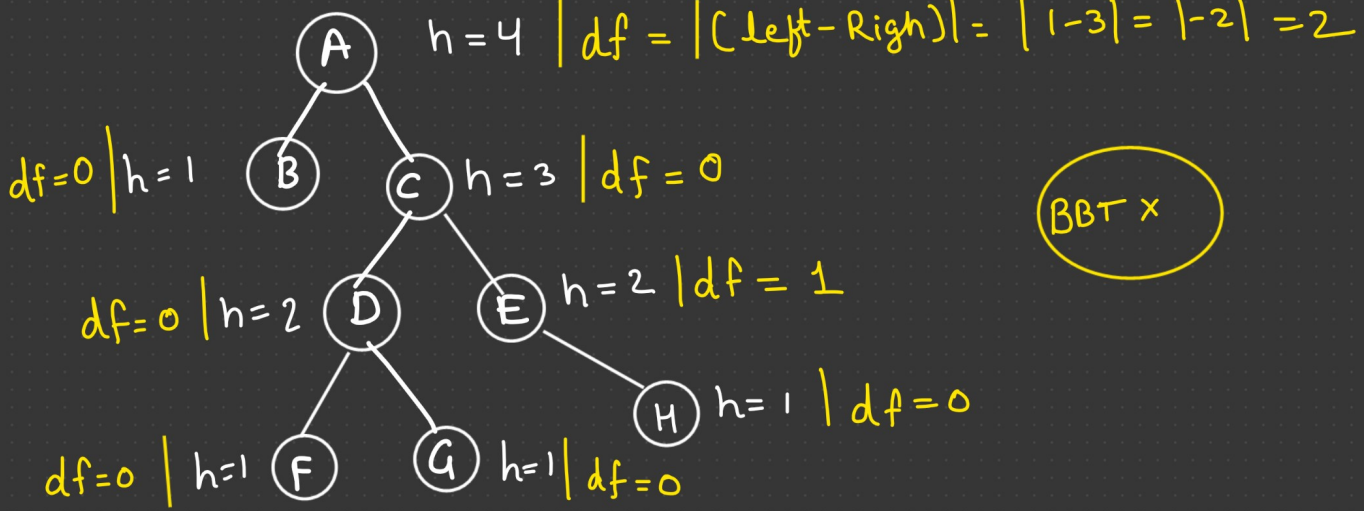


Balanced BT :-

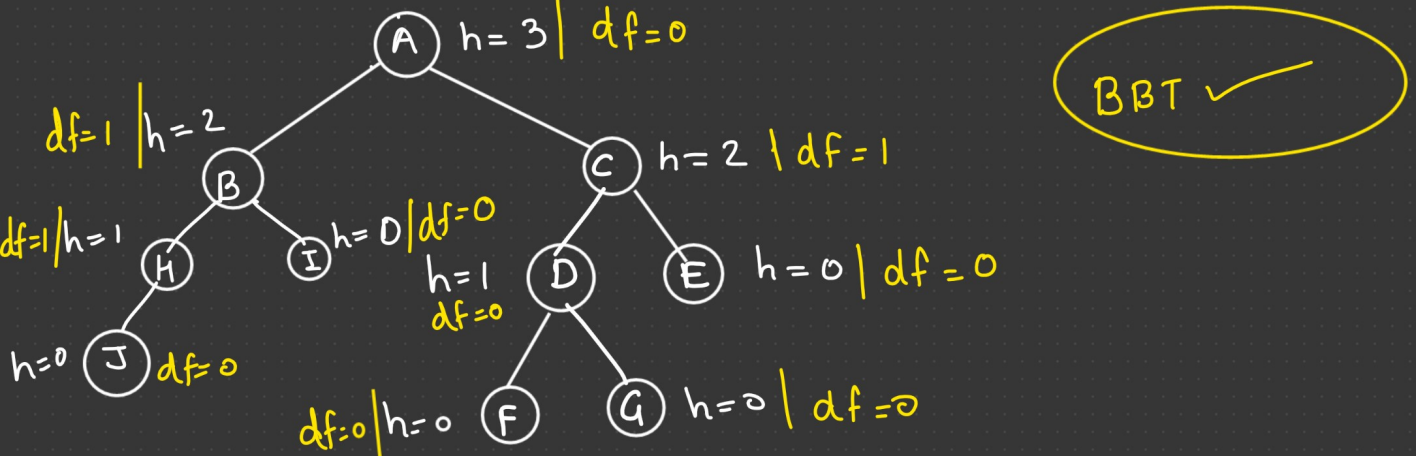
if (df = 0 'or' 1) ⟶ BBT



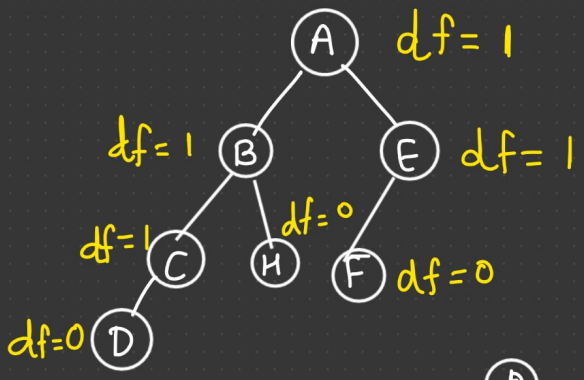
difference (df) = |(height of left subtree) − (height of right subtree)|



BBT ✗



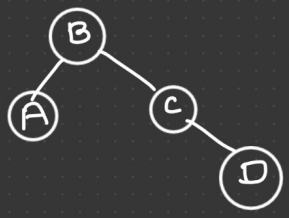
BBT ✓

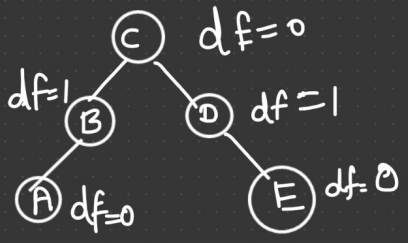

A
df = 1
df = 1
B
E
df = 1
df = 1
C
H
df = 0
F
df = 0
df = 0
D

→ BBT ✓


A
B
C
D


A
B
C
D


B
A
C
D


C
df = 0
df = 1
B
D
df = 1
A
df = 0
E
df = 0

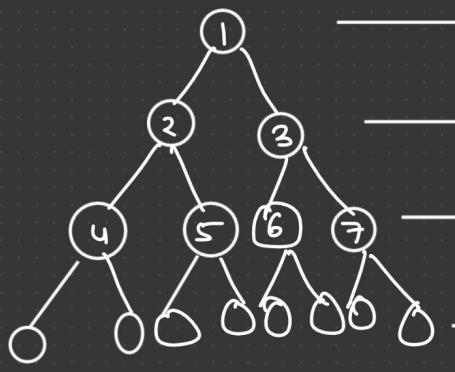
Properties of BT :-

1) The maximum number of nodes in a Binary Tree at level "l" is 2^l.

minimum

degenerated BT

* minimum no. of nodes in BT at any level is 1.



$\longrightarrow 1 \longrightarrow$ level = 0 $= 2^0$

$\longrightarrow 2 \longrightarrow$ level = 1 $= 2^1$

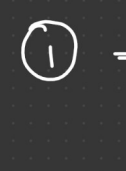
$\longrightarrow 4 \longrightarrow$ level = 2 $= 2^2$

$\longrightarrow 8 \longrightarrow$ level = 3 $= 2^3$

$\vdots$

level = l $= 2^l$

2) Minimum no. of nodes in a BT. of level l.



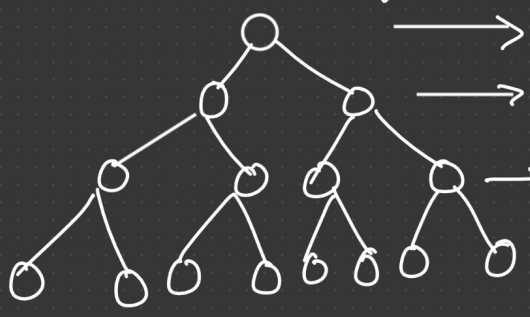
⇒ Level = 0
No. of nodes = 1
height = 1

level = 1
no. of nodes = 2
height = 2

level = 2
no. of nodes = 3
height = 3

Minimum no. of nodes (L+1) or (h).

3) Maximum no. of nodes in a BT.



$\longrightarrow l = 0 \Rightarrow 1 \Rightarrow 2^0$

$\longrightarrow l = 1 \Rightarrow 2 \Rightarrow 2^1$

$\longrightarrow l = 2 \Rightarrow 4 \Rightarrow 2^2$

$\longrightarrow l = 3 \Rightarrow 8 \Rightarrow 2^3$

$\vdots$

$\Rightarrow 2^l$

l = 2
h = 3
n = 7

Max node = $2^h - 1 = 2^{l+1} - 1$

$= 8 - 1 = 7$

Total no. of nodes $= 1+2+4+8 \ldots +2^{l}$

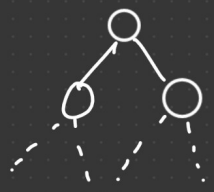
Sum of G.P $= \frac{a(r^n-1)}{(r-1)} = \frac{1.(2^{l+1}-1)}{(2-1)} = \boxed{2^{h}-1 = 2^{l+1}-1}$

4) $\boxed{e = n-1}$

5) No. of Null Reference = no. of nodes + 1.

$n = 1$
null = 2

$n = 2$
null = 3



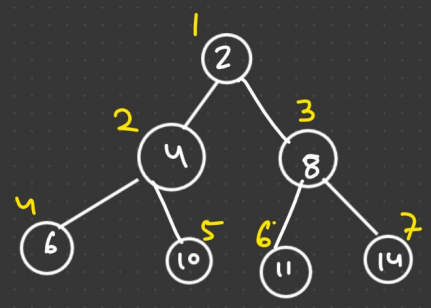
$n = 3$
null = 4

Binary Tree Representation :-

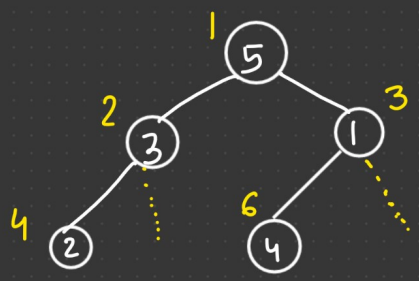
- 1) Array Representation
- 2) Linked list Representation

1) Array Representation :-

1 (Parent) = LC = 2 , RC = 3
2 (parent) = LC = 4 , RC = 5

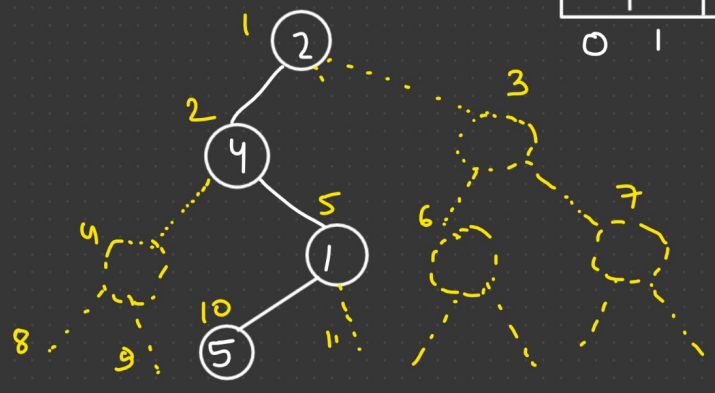


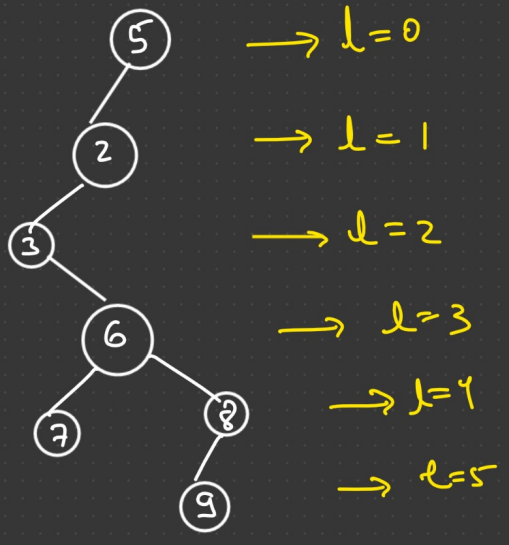
	2	4	8	6	10	11	14	
0	1	2	3	4	5	6	7	8



	5	3	1	2	–	4	–	
0	1	2	3	4	5	6	7	8

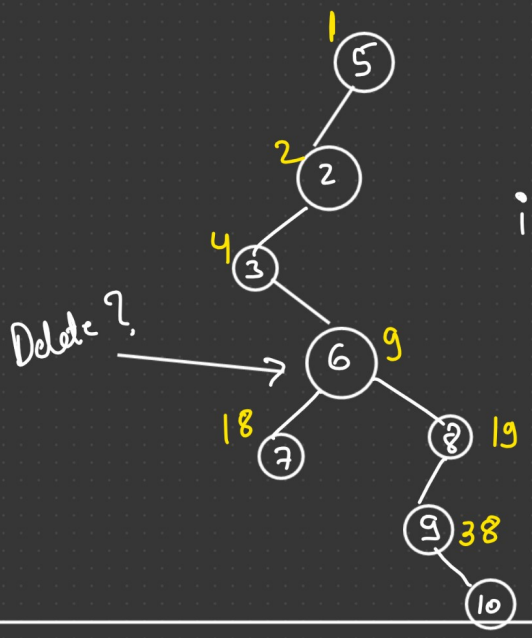
	2	4	–	–	1	–	–	–	–	5
0	1	2	3	4	5	6	7	8	9	10





$= 2^6 - 1 = 63$

$= 2^5 - 1 = 31$



7 elements ⇒ 38 nodes

if (K is parent node index)
then
2K = left child index
and
2K+1 = Right child index

⁂ K is index of child node then its parent node is at $\lfloor K/2 \rfloor$.

$\lfloor 3.5 \rfloor = 3$, $\lfloor 3.1 \rfloor = 3$, $\lfloor 4.9 \rfloor = 4$, $\lfloor 5 \rfloor = 5$

```
class Tree
{
   int a[100];

void insert(int v, int n)
   {
      a[n] = v;
   }

void insertleftchild(int v, int pi)
   {
      a[2*pi] = v;
   }
```

```
Void insertRightChild(int v, int pi)
{
    a[2*pi+1] = v;
}
```

Drawbacks of Array implementation :-

1) we can't change the size of Tree dynamically.

2) Wastage of memory.

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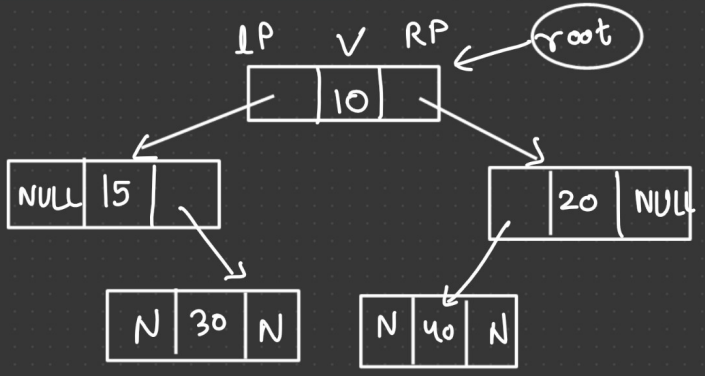
2) Linked List Reprentation :-



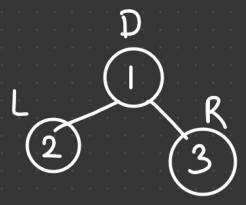

```
class node
{
 Public:
    int data;
    node *left;
    node *right;
    node(int v)
    {
      data = v;
      left = NULL;
      right = NULL;
    }
};
```

```
int main()
{
    node *root = new node(10);
    root->left = new node(15);
    root->right = new node(20);
    root->left->right = new node(30);
    root->right->left = new node(40);
}
```

This code can be used for testing any function.

Tree Traversal :-

L, R, D



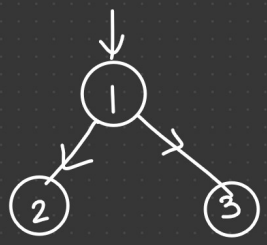
D, L, R
D, R, L → 1 (Root) → Preorder

L, D, R
R, D, L → 2 (Root) → Inorder

L, R, D
R, L, D → 3 (Root) → Postorder

→ Level order

1, 2, 3
1, 3, 2
2, 1, 3
2, 3, 1
3, 1, 2
3, 2, 1

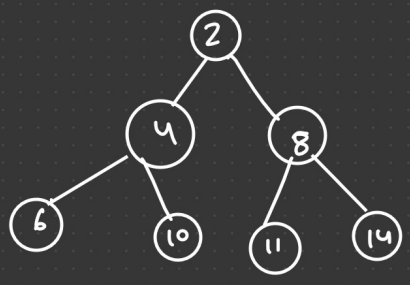


6

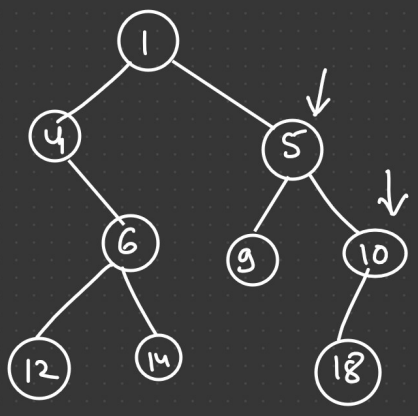
Preorder Traversal :-

(Root → left subtree → Right subtree) (It is same as DFS)

(we can use stack)

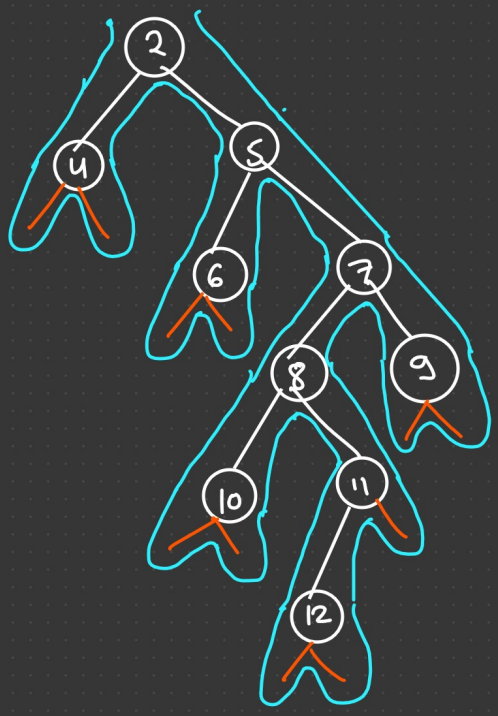


2, 4, 6, 10, 8, 11, 14



1, 4, 6, 12, 14, 5, 9, 10, 18 → Preorder

4, 12, 6, 14, 1, 9, 5, 18, 10 → Inorder.



2, 4, 5, 6, 7, 8, 10, 11, 12, 9 → Preorder

1st (Preorder) [D – L – R]

2, 4, 5, 6, 7, 8, 10, 11, 12, 9

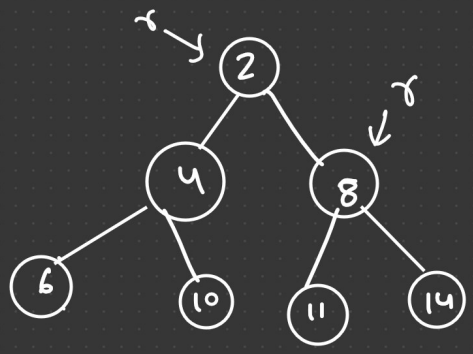
2nd (Inorder) [L – D – R]

4, 2, 6, 5, 10, 8, 12, 11, 7, 9

3rd (Postorder) [L – R – D]

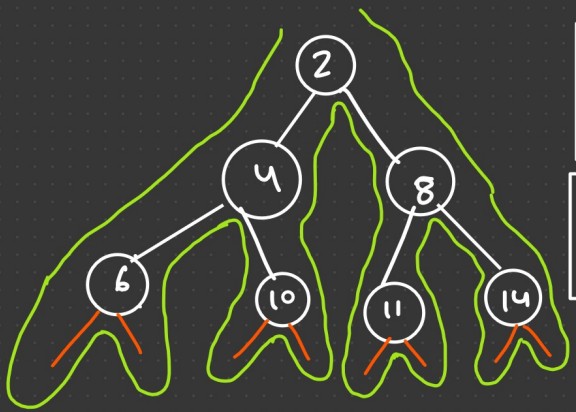
4, 6, 10, 12, 11, 8, 9, 7, 5, 2

Inorder Traversal :- (Left → Root → Right)



6, 4, 10, 2, 11, 8, 14

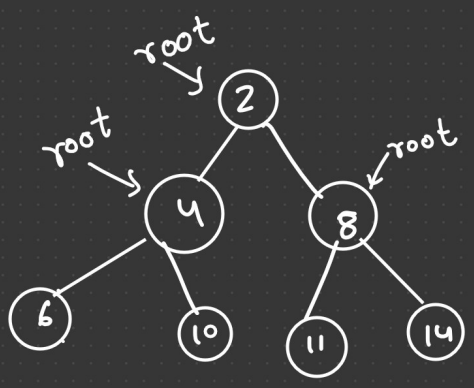
Post order Traversal :— (Left Subtree → Right Subtree → Data)



6, 10, 4, 11, 14, 8, 2 ← (3rd time) post order traversal.

6, 4, 10, 2, 11, 8, 14 ← Inorder (2nd time)

6, 4, 10, 2, 11, 8, 14



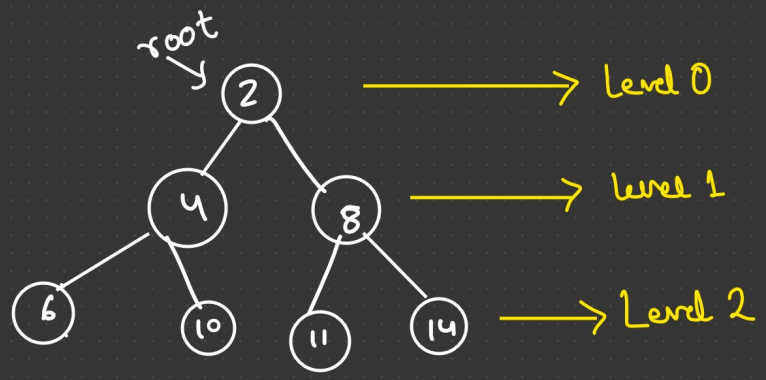
Preorder Traversal :-

[D → L → R]

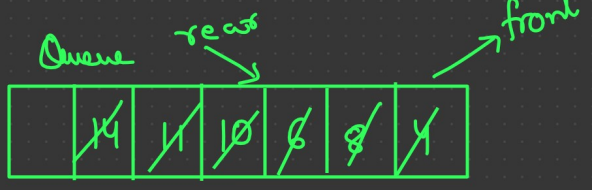
print Visit Visit

```
Void Preorder( Node * root)
{
    if ( root != NULL)
    {
        cout << root → data << " ";
        preorder (root → left);
        preorder (root → right);
    }
}
```

Level order Traversal :- (It is same as BFS)

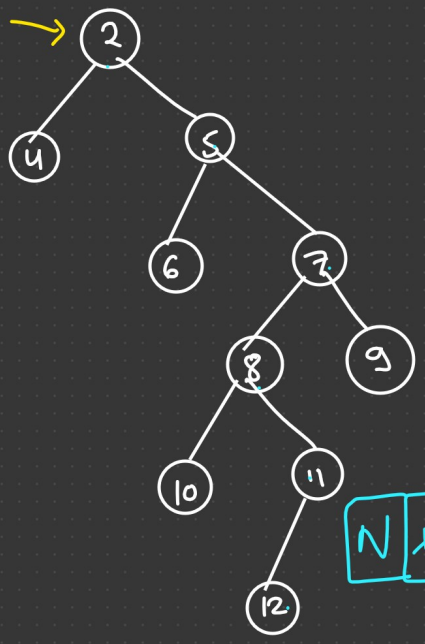


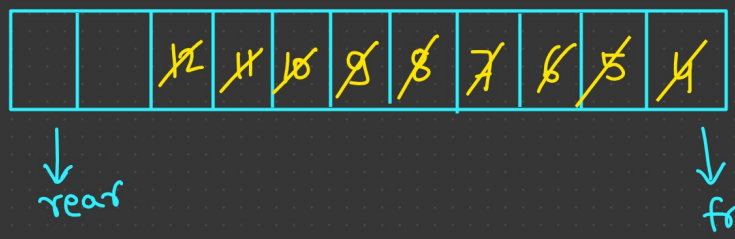
2, 4, 8, 6, 10, 11, 14



* we will use Queue DS to implement Level order Traversal.

2, 4, 8, 6, 10, 11, 14

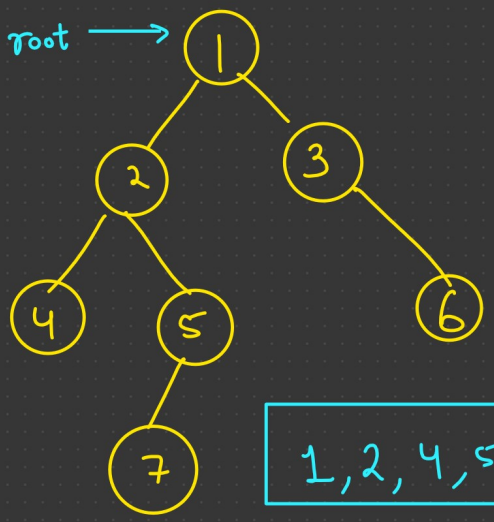




2, 4, 5, 6, 7, 8, 9, 10, 11, 12

count = 1 2 3 4 5 6

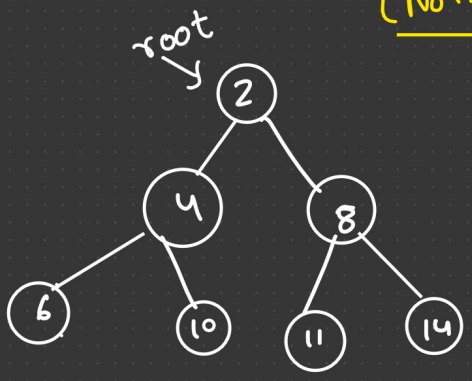
```
Void preorder ( Node *root)
{
    if( root != NULL)
    {
        cout<< root->data;
        preorder (root->left);
        preorder (root->right);
    }
}
int main()
{
  preorder (root);
}
```

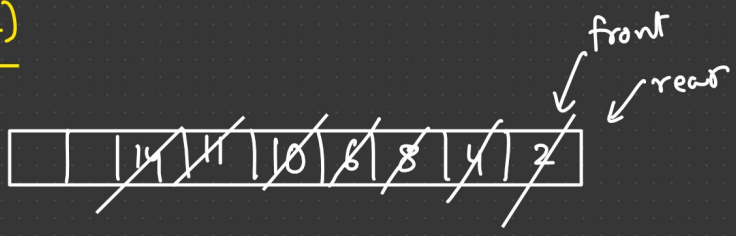


1, 2, 4, 5, 7, 3, 6

Level order Traversal :- (It is same as BFS)

(Non- Recursive)

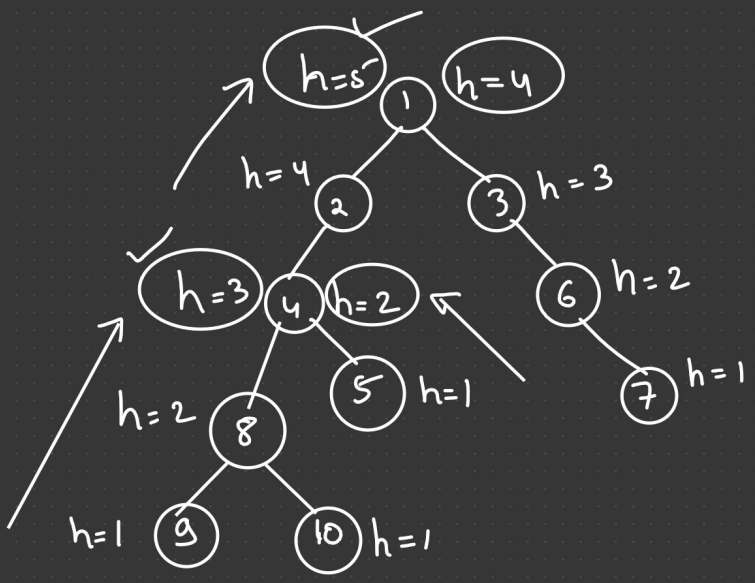




2, 4, 8, 6, 10, 11, 14,

```
Void printLevelorder(Node *root)
{
    if(root == NULL)
        return;
    queue<Node*> q;
    q.push(root);

    while( q.empty() == false)
    {
        Node *temp = q.front();
        cout<< temp -> data;
        q.pop();
        if( temp->left != NULL)
            q.push(temp->left);
        if( temp->right != NULL)
            q.push(temp->right);
    }
}
```

Height of Tree :- (Recursive)



Height of Tree ⇒ Height(1) = 5

Height(4) = 3

Height(1) = [lTree = Height(2) = 4

rTree = Height(3) = 3

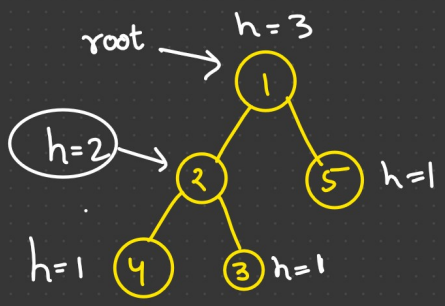
if(lTree > rTree)

return lTree + 1;

else

rTree + 1;]

```
int Height(Node *root)
{
    if(root == NULL)
        return 0;
    else {
        int lHeight = Height(root->left);
        int rHeight = Height(root->right);
        if(lHeight > rHeight)
            return lHeight + 1;
        else return rHeight + 1;
    }
}
```



H(1) = 3

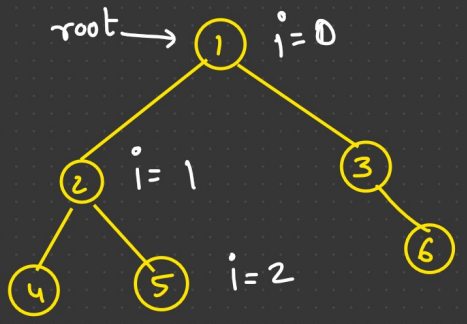
l = H(2) ⇒ 2

r = H(5) = 1

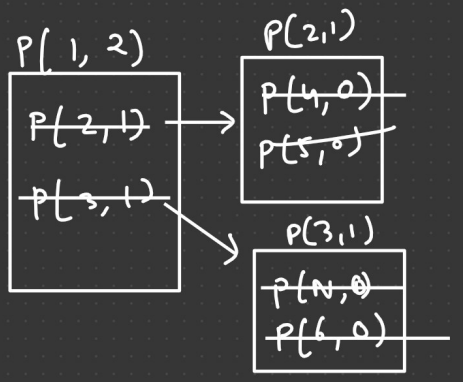
return 3

Level order Traversal (Recursive) :-

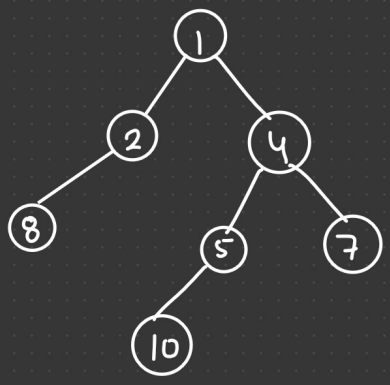
```
void printLevelOrder(Node *root)
{
    int h = Height(root);        = 3

    for(int i=0; i<h; i++)
        printCurrentLevel(root, i);
}

void printCurrentLevel(Node *root, int level)
{
    if(root == NULL)
        return;
    if(level == 0)
        cout << root->data << " ";
    else if(level > 0)
        printCurrentLevel(root->left, level-1);
        printCurrentLevel(root->right, level-1);
}
```



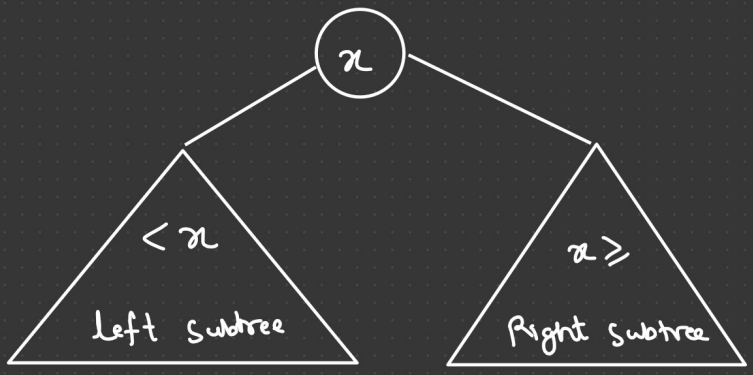
1, 2, 3, 4, 5, 6



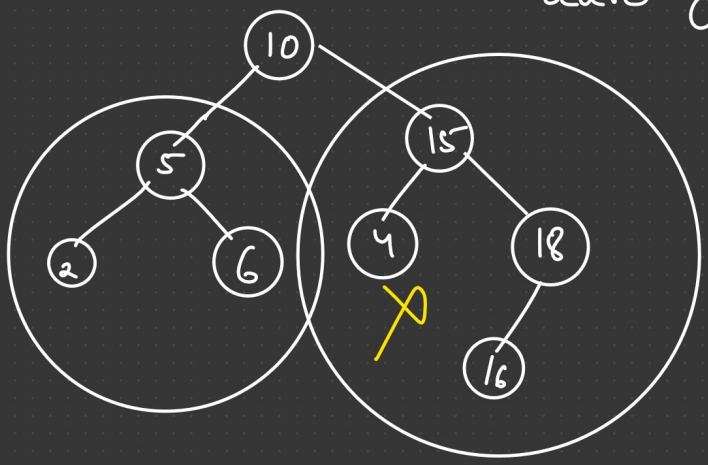
Binary Search Tree



Time Complexity :- $O(n)$

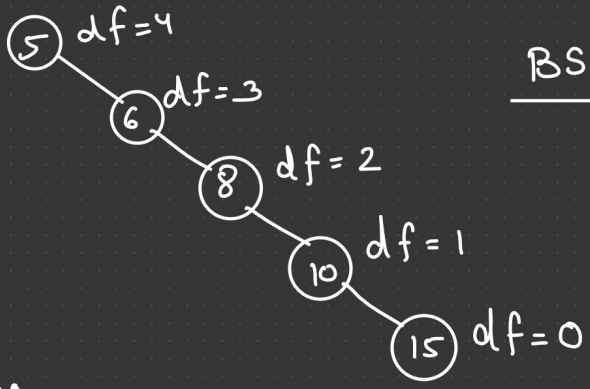


Searching TC = $O(\log n) < O(n)$



X BST

$$n + \frac{n}{2} + \frac{n}{4} + \dots + 1 = \log n$$



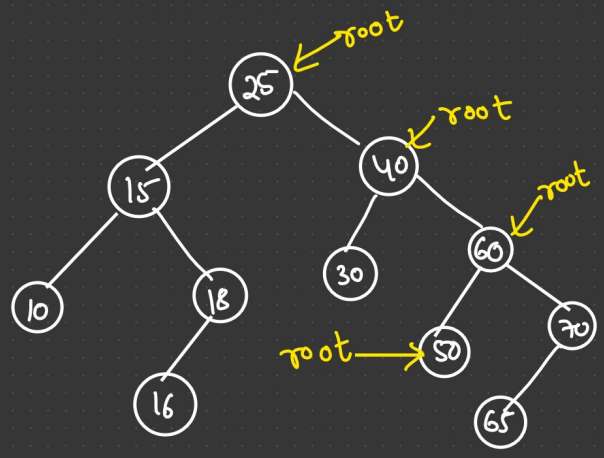
BST ✓

Problems:-

So, if BST is a skew tree then it will take $O(n)$ in searching.

Searching :—

Key = 50

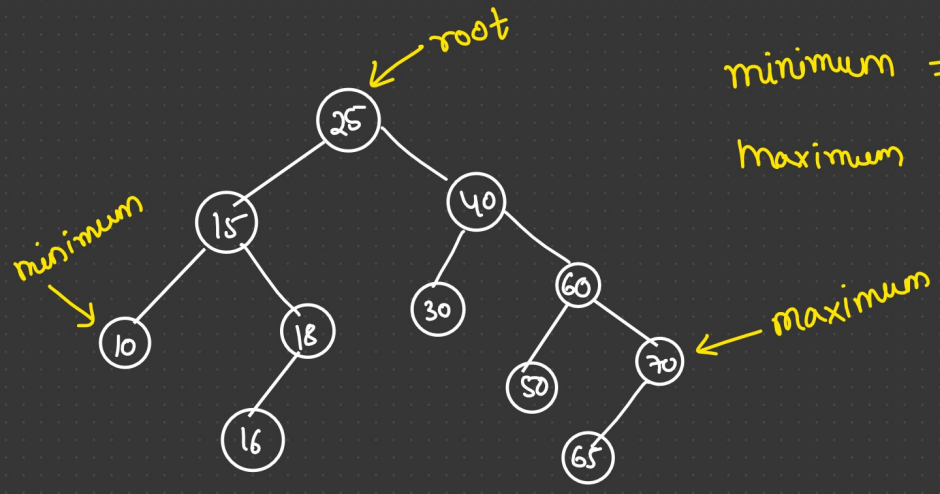



```
Node *Search(Node *root, int key)
{
  if(root == Null )
      return NULL;
  if(root->data == key)
      return root;
  else if (root->data > key)
  return search(root->left, Key);
  else
  return search(root->right, Key);
}
```

```
Class Node
{
   int data;
   Node *left;
   Node *right;
};
```

```
Node *root;
```

Minimum & Maximum value in a BST :-



minimum ⇒ Left most element of Tree

maximum ⇒ Right most element of Tree

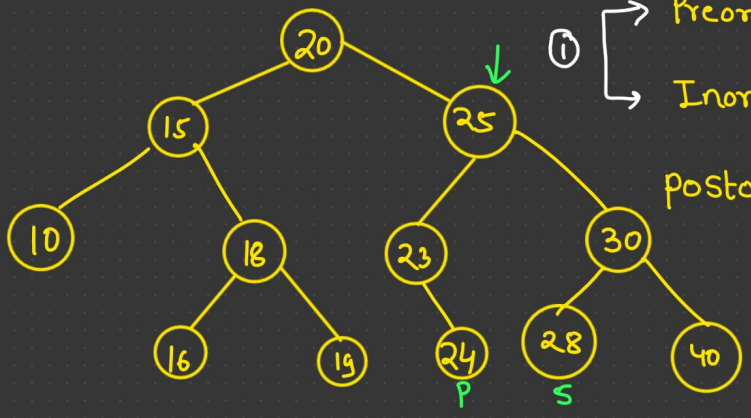
```
Node * MinElement ( Node *root)
{
    if (root == NULL)
        return NULL;
    else if ( root->left == = NULL)
        retuen root ;
    else
        minElement (root->left);
}
```

```
Node * temp = root;
if ( temp == NULL)
    return Null;
while ( temp->left != NULL)
{
  temp = temp -> left;
}
  return temp ;
```

```
Node * MaxElement ( Node * root )
{
    if (root == NULL)
        return NULL;
    else if ( root->right == = NULL)
        retuen root ;
    else
        maxElement (root-> right) ;
}
```

```
Node * temp = root;
if ( temp == NULL)
    return Null;
while ( temp -> right != NULL)
{
  temp = temp -> right;
}
  return temp ;
```

Inorder Successor & Inorder predecessor of BST :-



Preorder = 20, 15, 10, 18, 16, 19, 25, 23, 24, 30, 28, 40

Inorder = 10, 15, 16, 18, 19, 20, 23, 24, 25, 28, 30, 40

postorder = 10, 16, 19, 18, 15, 24, 23, 28, 40, 30, 25, 20

Inorder = 10, 15, 16, 18, 19, 20, 23, 24, 25, 28, 30, 40

Inorder Successor of (E) = Minimum Element of its Right Subtree.

Inorder Predecessor of (E) = Maximum Element of its left Subtree.

[Node → Right → left most] (1 step)

[Node → Left → Right most] (1 step)

```
Node * InorderSuccessor (Node * node)
{

    return MinElement (node → right);

}
```

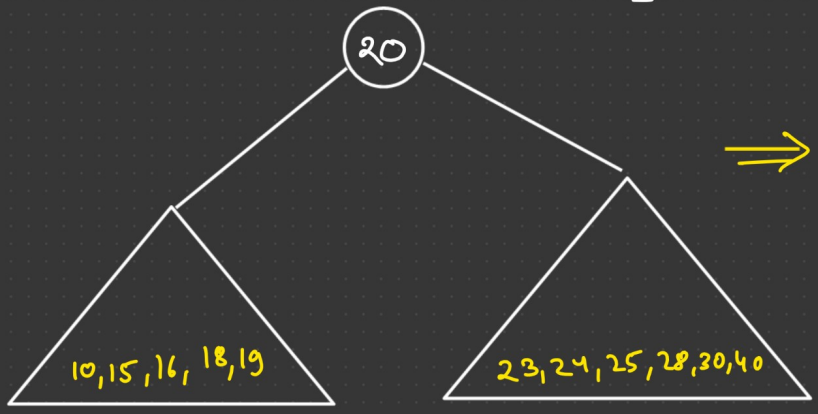
```
Node *InorderPredecessor (Node * node)
{
    return MaxElement (node → left);
}
```

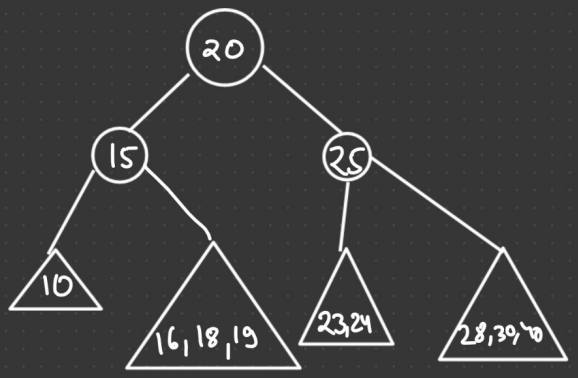
Create BST using Inorder & preorder Traversal :-

Preorder = 20, 15, 10, 18, 16, 19, 25, 23, 24, 30, 28, 40

Inorder = 10, 15, 16, 18, 19, 20, 23, 24, 25, 28, 30, 40

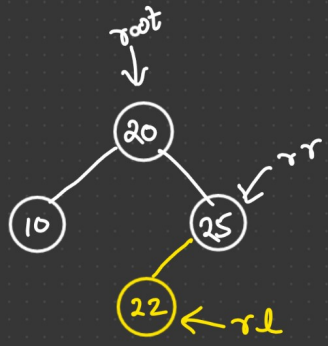
Postorder & Inorder ✓

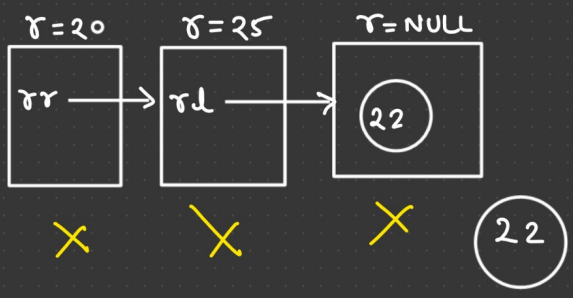




Preorder & Postorder = ? (Not possible)

Creation / Insertion in BST :-



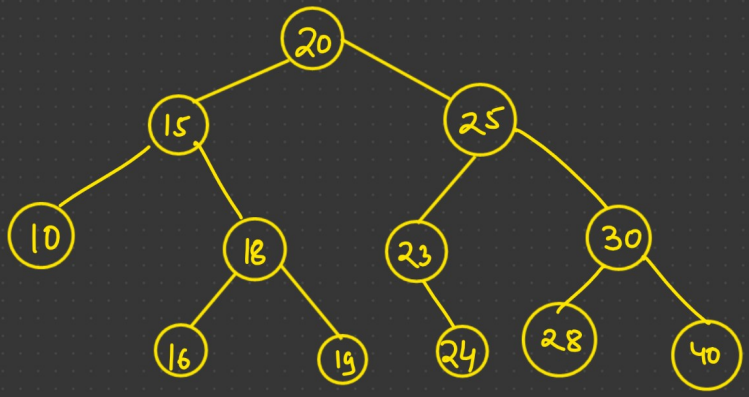



```
Node * Insert(Node *root, int value)
{
    if (root == NULL)
        return new Node(value);
    if (root -> data < value)
    root->right = Insert(root->right, value);
    if (root -> data >= value)
    root->left = Insert(root->left, value);
}
```

1) Time complexity of Inserting a node = O(logn)

2) Time complexity to create BST = O(n logn)

Deletion in BST :-

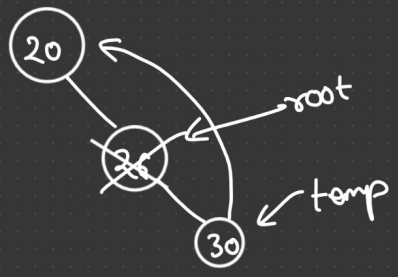


1) Leaf Node (zero child)

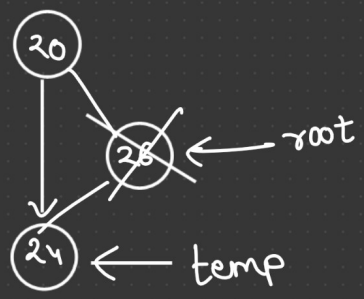
2) Non-Leaf Node (one child)

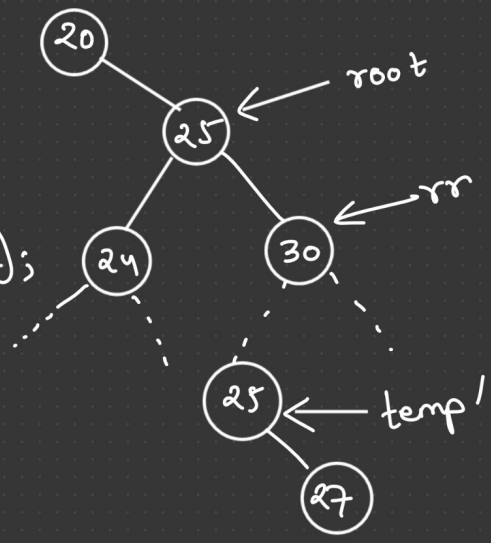
3) —•— (two child)

```
Node * delete( Node * root, int value)
{
    if( root == NULL)
        return root;

  if( root->data < value)
   root->right = delete( root->right, value);
  else if( root->data > value)
   root->left = delete( root->left, value);

  else { if ((root->left == NULL) && (root->right == NULL))  //zero child
     {  delete root;
        return NULL;
     }
    else if( root->left == NULL)
    {
      Node * temp = root->right;
        delete root;
        return temp;
    }
```



```
else if (root->right == NULL)
{
    Node *temp = root->left;
    delete root;
    return temp;
}
else{          // two child
    Node* temp = minElement(root->right);
    root->data = temp->data;
    root->right = delete(root->right, temp->data);
}
return root;
}
```





AVL Tree :-